

GREAT DUN FELL NO2, United Kingdom

WMO: 032270

Lat: 54.6842N

Lon: 2.4514W

Elev: 847

StdP: 91.56

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.2	-6.1	-14.0	1.2	3.1	-9.3	1.9	-2.7	46.3	0.2	40.9	-0.2	12.8	340	1.296

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.5	17.8	13.5	16.1	12.8	14.5	12.0	14.8	16.2	13.8	15.2	12.8	14.0	7.1	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.3	11.3	15.1	13.3	10.5	14.0	12.4	9.9	13.2	44.4	16.4	41.5	15.3	38.8	14.1	28.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
37.7	31.4	26.0	-8.4	21.7	2.0	3.0	-9.9	23.8	-11.1	25.6	-12.2	27.2	-13.7	29.4	
			DB												
			WB	-8.8	18.0	1.5	3.6	-9.8	20.6	-10.7	22.8	-11.5	24.8	-12.6	27.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	4.4	-0.3	-0.1	0.9	3.2	5.7	8.0	10.1	9.8	7.9	5.1	2.2	0.0
	DBStd	4.79	2.83	3.06	3.14	3.44	3.44	2.83	2.84	2.52	2.47	2.79	2.98	3.12
	HDD10.0	2136	319	282	283	207	141	73	33	32	71	152	233	308
	HDD18.3	5081	577	516	541	455	391	309	257	264	312	409	483	567
	CDD10.0	97	0	0	0	2	8	14	36	27	9	1	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	1	0	0	0	0	0	0	1	1	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	12.2	14.9	14.8	13.4	11.0	10.3	9.6	9.5	10.5	10.6	13.5	14.2	14.2
Precipitation	PrecAvg	1367	161	122	97	81	81	79	90	102	103	128	148	173
	PrecMax	1699	255	296	183	150	149	178	201	245	190	250	368	450
	PrecMin	997	33	22	29	37	22	25	49	22	26	47	69	53
	PrecStd	179	61	75	34	31	35	39	34	45	39	50	67	85
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.6	9.5	11.3	14.3	17.7	18.6	20.2	19.8	16.4	13.1	11.2	9.4
		MCWB	2.9	5.1	5.9	8.6	12.9	12.9	15.2	14.3	14.1	10.3	7.6	4.3
	2%	DB	6.2	6.8	8.0	11.6	14.9	16.0	17.7	17.4	14.0	11.5	9.1	6.9
		MCWB	5.7	5.0	5.6	8.4	10.8	12.0	13.4	13.8	12.5	10.3	8.4	6.6
	5%	DB	4.9	5.3	6.5	9.7	12.8	13.8	16.1	14.9	12.6	10.3	7.7	5.6
		MCWB	4.6	4.5	5.2	7.4	9.7	11.4	13.0	12.9	11.3	9.4	7.3	5.3
	10%	DB	3.8	4.1	5.1	7.9	10.7	12.2	14.3	13.3	11.4	9.2	6.4	4.6
		MCWB	3.7	3.5	4.4	6.6	8.4	10.5	12.3	12.1	10.4	8.6	6.1	4.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.7	7.7	7.7	10.9	14.2	15.5	16.9	15.7	14.9	12.1	9.8	8.0
		MCDB	6.8	8.1	8.2	12.0	16.3	16.8	18.2	17.5	15.8	12.3	9.8	8.3
	2%	WB	5.5	5.8	6.5	9.2	12.1	13.0	14.9	14.4	13.2	10.7	8.4	6.6
		MCDB	5.6	6.1	7.4	10.5	14.4	14.6	16.2	16.1	13.7	10.9	8.5	6.6
	5%	WB	4.3	4.7	5.5	7.9	10.2	11.8	13.9	13.4	11.8	9.6	7.3	5.3
		MCDB	4.4	5.0	6.2	9.2	11.9	13.0	15.2	14.7	12.3	9.8	7.3	5.4
	10%	WB	3.1	3.7	4.4	6.9	9.0	10.7	12.7	12.5	10.7	8.6	6.1	4.3
		MCDB	3.2	4.0	4.9	8.0	10.3	11.9	14.0	13.4	11.2	8.9	6.2	4.4
Mean Daily Temperature Range	5% DB	MDBR	3.6	3.7	3.9	4.5	4.9	4.5	4.5	4.0	3.9	3.6	3.6	3.6
		MCDBR	5.4	5.2	5.4	6.9	7.0	6.5	6.6	6.2	4.9	5.2	5.3	5.2
		MCWBR	5.1	4.0	4.5	5.0	5.6	5.3	5.2	4.7	4.2	5.3	5.3	4.9
	5% WB	MCDBR	5.1	4.3	4.8	5.2	6.3	5.9	5.8	5.2	4.6	4.9	5.1	5.0
		MCWBR	5.0	4.3	4.6	4.5	5.5	5.3	5.1	4.5	4.5	5.1	5.2	5.0
Clear-Sky Solar Irradiance	taub	0.286	0.299	0.323	0.341	0.343	0.342	0.343	0.333	0.318	0.306	0.284	0.269	
	taud	2.337	2.436	2.390	2.356	2.383	2.399	2.395	2.436	2.487	2.496	2.408	2.295	
	Ebn at Noon	667	802	856	886	902	904	896	885	855	783	671	598	
	Edh at Noon	54	69	90	107	110	110	109	98	82	64	50	45	
All-Sky Solar Radiation	RadAvg	0.55	1.22	2.32	3.71	4.82	4.84	4.67	3.77	2.68	1.44	0.70	0.42	
	RadStd	0.07	0.14	0.22	0.37	0.36	0.47	0.49	0.28	0.23	0.17	0.07	0.04	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (57 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page